

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***First Semester M.Tech. (VDE) (Credit System) Degree Examinations****Supplementary Examinations - September - October 2021****20VDE101 – DIGITAL VLSI DESIGN**

Duration: 3 Hours

Max. Marks: 100

Note: Answer any Five full questions.

	Marks	BT*	CO*	PO*
1. a) Illustrate the electrical behavior of the MOS structure having three distinct regions of operation when an external bias is applied. Show corresponding energy band diagrams.	8	L*2	1	3
b) A n-channel MOSFET has the following voltage and current data. $V_{GS}=5V$, $V_{DS}=5V$, $V_{SB}=3V$ and $I_D=347 \mu A$. Calculate V_T and γ . Assume $\phi_F = -0.3V$, $K_n=60 \mu A/V^2$, $V_{TO}=1.2V$.	4	L3	1	3
c) Analyze the working of a depletion - load nMOS Inverter. Drive expressions for V_{OH} , V_{OL} , V_{IL} and V_{IH} .	8	L2	1	3
2. a) With neat diagram and appropriate explanation analyse the channel length modulation effect. Drive an expression for drain current due to channel length modulation.	8	L3	1	3
b) Briefly explain the following. i) Hot carrier effect ii) Sub threshold conduction	5	L2	1	3
c) Consider a resistive load inverter circuit with $V_{DD}=5V$, $K'_n = 20 \mu A/V^2$, $V_{TO} = 0.8V$, $R_L = 200K\Omega$ and $W/L = 2$. Calculate V_{IH} of the VTC and find noise margin for high of the circuit.	7	L3	1	3
3. a) Consider a CMOS Inverter circuit with the following parameters. $V_{DD} = 3.3V$, $V_{TO,n} = 0.6V$, $V_{TO,p} = -0.7V$, $K_n = 200 \mu A/V^2$, $K_p = 80 \mu A/V^2$. Calculate Noise margins.	8	L3	2	3
b) With neat diagram derive expression to calculate delays T_{pHL} and T_{pLH} of a CMOS Inverter.	12	L2	2	3
4. a) With neat diagram illustrate the switching power dissipation of CMOS Inverter.	10	L2	2	3
b) In a CMOS inverter, $V_{DD}=5V$, Output load capacitance = 1pF, $\mu_n C_{ox} = 20 \mu A/V^2$, $(W/L)_n = 10$, $V_{T,n}=1.0V$. Calculate τ_{fall} corresponding to time elapsed for the output to fall from $V_{90\%} = 4.5V$ to $V_{10\%}=0.5V$ using i) Average current method ii) Differential equation method.	10	L3	2	3
5. a) Design a 4:1 Mux using minimum number of CMOS Transmission gates.	6	L3	3	3
b) Evaluate the working of a Dynamic CMOS Logic.	8	L2	3	3
c) Implement the Logic $Y = \overline{A + BC + DE}$ using BICMOS Logic.	6	L3	3	3

P.T.O.

6.	a)	Examine the concept of voltage Bootstrapping used in dynamic circuits.	6	L2	3	3
	b)	With an example explain how Domino CMOS Logic can overcome the problems faced by the CMOS Dynamic Logic.	8	L2	3	3
	c)	Implement the Logic $Y=A+BC$ using minimum number of pass transistors.	6	L3	3	3
7.	a)	With the help of neat diagram explain the working of a bit-serial adder.	6	L3	4	3
	b)	With the help of neat diagram illustrate zero/one detectors.	6	L3	4	3
	c)	Outline the working of a 4 bit barrel shifter with the help of neat circuit diagram.	8	L3	4	3
8.	a)	Demonstrate the working of a 2x2 array multiplier.	6	L3	4	3
	b)	Discuss how a comparator can be designed using address.	6	L3	4	3
	c)	Examine the working of a Transmission gate adder using appropriate circuit diagrams.	8	L3	4	3
9.	a)	Discuss on-chip clock generation and distribution systems available for a digital system.	10	L2	5	3
	b)	Discuss the following terms used in design for manufacturability				
		i) Circuit parameters				
		ii) Noise parameter distribution	10	L2	5	3
10.	a)	With appropriate diagrams explain the concept of Latch-up in CMOS IC's. Discuss some of the prevention techniques.	10	L2	5	3
	b)	Elaborate on ESD protection available in CMOS IC's.	10	L2	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
